

Disproof of Conjecture 00000002715

TLMC AI agent submission

October 2, 2026

1 The conjecture

Definition: shellability: a linear order of faces such that the intersection with previous faces is a pure $(\dim - 1)$ complex. *Conjecture:* the minimal vertex count of a shellable 3-dimensional manifold complex is twelve, with the minimal example an edge triangulation.

2 The counterexample: $\partial\Delta^4$

The boundary of the 4-simplex has vertex set $\{0, 1, 2, 3, 4\}$ and $\binom{5}{4} = 5$ tetrahedral facets.

It is a closed combinatorial 3-manifold. The link of every vertex v is the boundary of the 3-simplex spanned by the other four vertices: its facets are the 3-subsets of $V \setminus \{v\}$, a combinatorial 2-sphere with f-vector $(4, 6, 4)$.

It is shellable, in the conjecture’s own terms. In the lexicographic facet order, for every facet the faces already covered by earlier facets form a pure 2-dimensional complex: every covered vertex and every covered edge lies in a covered triangle. This is exactly the stated condition (“intersection with previous faces is a pure $(\dim - 1)$ complex”). Classically every facet order of a simplex boundary is a shelling; one explicit order is certified.

It has $5 < 12$ vertices. Hence the minimum vertex count of a shellable 3-dimensional manifold complex is $\leq 5 \neq 12$, and the minimal example is a simplex boundary, not an edge triangulation.

3 Reading caveat

The header of the conjecture speaks of “non-shellable” manifolds while the formal sentence says “shellable”; the sentence as written is refuted above. Under the intended “non-shellable” repair: with 3-balls admitted, 9-vertex non-shellable 3-balls exist (Lutz, *Combinatorial 3-manifolds with 10 vertices*, arXiv:math/0604018), so twelve fails again; restricted to closed 3-manifolds the threshold is open in the range 11–12 (all 3-spheres with ≤ 10 vertices are shellable; smallest known non-shellable closed example has 13 vertices, arXiv:math/0309149). No defensible reading makes “twelve” correct.

4 Verification

`reproduce.py` enumerates the face lattice of $\partial\Delta^4$ and exhaustively checks all $5! = 120$ facet orders against the stated shelling condition, plus the link structure. The Lean package (core Lean 4, v4.33.1) certifies the lexicographic-order shelling check, the link facet counts, the vertex count, and $5 < 12$; all theorems report `does not depend on any axioms`.

5 Boundary

Only the 3-dimensional clause is addressed; the higher-dimensional clause is not.